

Questions/Answers

Part 1: Examine the Network and Evaluate the Need for Static Routing

- a. Looking at the topology diagram, how many networks are there in total?

Ans: 6 networks (3 LANs and 3 WAN links)

- b. How many networks are directly connected to R1, R2, and R3?

Ans: R1: 2 networks

R2: 3 networks

R3: 2 networks

- c. How many static routes are required by each router to reach networks that are not directly connected?

Ans: R1: 3 routes

R2: 2 routes

R3: 3 routes

- d. Test connectivity to the R2 and R3 LANs by pinging PC2 and PC3 from PC1.

Why were you unsuccessful?

Ans: Because no static routes are configured yet, so routers don't know paths to remote networks.

Part 2: Configure Static and Default Routes

Step 1: Configure recursive static routes on R1.

- a. What is recursive static route?

Ans: A route that specifies only the next-hop IP address without an exit interface.

- b. Why does a recursive static route require two routing table lookups?

Ans: First lookup finds the next-hop IP, second lookup finds the exit interface to reach that IP.

- c. Configure a recursive static route to every network not directly connected to R1, including the WAN link between R2 and R3.

Ans: R1(config)# ip route 172.31.0.0 255.255.255.0 172.31.1.193

R1(config)# ip route 172.31.1.128 255.255.255.192 172.31.1.193

R1(config)# ip route 172.31.1.196 255.255.255.252 172.31.1.193

- d. Test connectivity to the R2 LAN and ping the IP addresses of PC2 and PC3.

Why were you unsuccessful?

Ans: R2 doesn't have return routes configured to reach R1's LAN.

Step 2: Configure directly attached static routes on R2.

- a. How does a directly attached static route differ from a recursive static route?

Ans: Uses exit interface instead of next-hop IP address.

- b. Configure a directly attached static route from R2 to every network not directly connected.

Ans: R2(config)# ip route 172.31.1.0 255.255.255.128 S0/0/0

R2(config)# ip route 172.31.1.128 255.255.255.192 S0/0/1

- c. Which command only displays directly connected networks?

Ans: show ip route connected

d. Which command only displays the static routes listed in the routing table?

Ans: show ip route static

e. When viewing the entire routing table, how can you distinguish between a directly attached static route and a directly connected network?

Ans: Directly connected shows "C" and interface. Static route shows "S" and may show interface or next-hop IP.

Step 3: Configure a default route on R3.

a. How does a default route differ from a regular static route?

Ans: Matches all networks not in routing table (0.0.0.0/0), while regular static route matches specific network.

b. Configure a default route on R3 so that every network not directly connected is reachable.

Ans: R3(config)# ip route 0.0.0.0 0.0.0.0 172.31.1.197

c. How is a static route displayed in the routing table?

Ans: Static routes appear with an "S", and a default route appears as "S*" 0.0.0.0/0.

Step 4: Document the commands for fully specified routes

a. Explain a fully specified route.

Ans: A fully specified static route includes both the exit interface and the next-hop IP address in a single command. This eliminates the need for recursive lookups and provides the most detailed path information to the destination network.

b. Which command provides a fully specified static route from R3 to the R2 LAN?

Ans: ip route 172.31.0.0 255.255.255.0 S0/0/1 172.31.1.197

c. Write a fully specified route from R3 to the network between R2 and R1. Do not configure the route; just calculate it.

Ans: ip route 172.31.1.192 255.255.255.252 S0/0/1 172.31.1.197

d. Write a fully specified static route from R3 to the R1 LAN. Do not configure the route; just calculate it.

Ans: ip route 172.31.1.0 255.255.255.128 S0/0/1 172.31.1.197

Step 5: Verify static route configurations.

Use the appropriate show commands to verify correct configurations. Which show commands can you use to verify that the static routes are configured correctly?

Ans: show ip route

show ip route static

show running-config